PAPER ID-311548

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2025-26
CONCRETE TECHNOLOGY

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO
a.	Enlist the composition of cement.	1
b.	Define admixtures and state their purpose in concrete.	2
c.	What is segregation in concrete?	3
d.	Define mix proportioning of concrete.	4
e.	Mention two applications of 3D printed concrete in construction.	4
f.	What is self-compacting concrete?	5
g.	What is recycled aggregate concrete?	5

SECTION B

2. Attempt any *three* of the following: **7 x 3 = 21**

a.	Describe the effects of shape and texture of aggregate for the strength and workability of the concrete.	1
b.	Discuss the effects of accelerators and retarders on concrete properties.	2
c.	Describe bleeding and laitance in concrete and their effects.	3
d.	Explain the step-by-step procedure of concrete mix design as per IS method.	4
e.	What are the advantages and disadvantages of using ready mix concrete instead of site mixed concrete?	5

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the setting and hardening of cement. Discuss the factors affecting setting time.	1
b.	State the requirements of water for mixing and curing of concrete.	1

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the production, properties, and applications of fly ash in concrete.	2
b.	Describe Metakaolin. Discuss advantages and disadvantages.	2

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the Vee-Bee consistometer test and its suitability for stiff concrete.	3
b.	Explain different methods of batching of concrete.	3

6. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Discuss the applications, advantages, and challenges of 3D printed concrete.	4
b.	Explain the concept of rheology of fresh concrete.	4

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7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Discuss the properties and applications of steel fibre reinforced concrete.	5
b.	Explain the present status, challenges, and future prospects of advanced concretes in India.	5

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